Cultural Bias Matters: A Cross-Cultural Benchmark Dataset and Sentiment-Enriched Model for Understanding Multimodal Metaphors

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Abstract

Automatic metaphor processing largely depends on English-centric data, risking cultural biases and overestimated generalization, especially in multimodal settings. In this work, we introduce **MultiMM**, a cross-cultural multimodal metaphor benchmark containing 8,461 text-image advertisements from Chinese and English sources with annotations for metaphor occurrence, source/target domains, and sentiment labels. We evaluate 18 models spanning text-only, vision-only, and multimodal approaches, revealing significant performance gaps between cultures. To address these challenges, we propose **SEMD** (Sentiment-Enriched Metaphor Detection), which integrates sentiment-aware representations into multimodal fusion, achieving state-of-the-art results: 80.16% F1 for English and 77.79% F1 for Chinese metaphor detection. Our analysis reveals that sentiment provides a universal bridge for cross-cultural metaphor understanding, with positive sentiment correlating more strongly with metaphors in English (74.75%) than Chinese (56.42%) advertisements. This work contributes the first cross-cultural multimodal metaphor dataset and demonstrates practical improvements toward culturally robust metaphor processing.

1 Introduction

Metaphors are fundamental to human communication, serving as cognitive tools that help us understand abstract concepts through concrete experiences (Shutova et al., 2010; Forceville and Urios-Aparisi, 2009). In the digital age, multimodal metaphors—combining text and visual elements—have become increasingly prevalent in advertising, social media, and online discourse. However, automatic metaphor processing systems exhibit significant cultural biases, as they are predominantly trained on English-centric datasets and evaluated within Western cultural contexts (Shutova et al., 2017; Forceville, 2017).

The challenge becomes particularly acute in multimodal settings, where cultural differences in visual symbolism and textual expression converge. For instance, while a dragon symbolizes power and prosperity in Chinese culture, it often represents danger or evil in Western contexts (Forceville, 2002). Such cultural variations in metaphorical understanding can lead to misinterpretation and reduced effectiveness of cross-cultural communication technologies.

Despite growing interest in multimodal metaphor processing, existing datasets and benchmarks remain culturally homogeneous. Most metaphor detection systems are developed and evaluated primarily on English data, limiting their generalizability across cultures (Shutova et al., 2013). This creates a critical gap: how can we build metaphor processing systems that are robust across different cultural contexts?

To address this challenge, we present three key contributions:

First, we introduce **MultiMM**, the first large-scale cross-cultural multimodal metaphor benchmark. Our dataset contains 8,461 text-image advertisements spanning Chinese (4,397 samples) and English (4,064 samples) sources, annotated for metaphor occurrence, source/target domains, and sentiment categories. This balanced representation enables systematic evaluation of cultural bias in metaphor processing systems.

Second, we propose **SEMD** (Sentiment-Enriched Metaphor Detection), a novel multimodal approach that leverages sentiment as a universal bridge for cross-cultural metaphor understanding. SEMD integrates sentiment-aware representations with visual and textual features, achieving state-of-the-art performance on both Chinese and English metaphor detection.

Third, we conduct comprehensive experiments with 18 baseline models across text-only, vision-only, and multimodal categories, revealing signif-

icant performance gaps between cultures and providing insights into the role of sentiment in cross-cultural metaphor processing.

Our analysis reveals fascinating cultural patterns: positive sentiment correlates more strongly with metaphorical content in English advertisements (74.75%) compared to Chinese ones (56.42%), highlighting fundamental differences in how cultures employ emotional appeals in metaphorical communication.

2 Related Work

2.1 Metaphor Detection

Computational metaphor processing has evolved from rule-based approaches to sophisticated neural architectures. Early work focused on identifying linguistic patterns and selectional preferences (Shutova et al., 2010), while recent advances leverage pretrained language models like BERT for improved metaphor detection (Choi et al., 2021; Devlin et al., 2019).

Cross-linguistic metaphor processing remains challenging due to cultural variations in metaphorical expressions (Shutova et al., 2017). Most existing work focuses on English, with limited exploration of how metaphor processing systems generalize across cultures and languages.

2.2 Multimodal Metaphor Understanding

Multimodal metaphors, where meaning emerges from the interaction between visual and textual elements, represent a crucial frontier in computational semantics (Forceville and Urios-Aparisi, 2009). Forceville's seminal work on pictorial metaphor established the theoretical foundation for analyzing visual metaphors in advertising (Forceville, 2002).

Recent computational approaches have begun incorporating visual features for metaphor detection (Shutova et al., 2016), but most focus on single-modal or limited multimodal scenarios. The integration of multiple modalities for metaphor processing remains an active area of research, particularly in cross-cultural contexts (Forceville, 2021).

2.3 Sentiment and Cultural Context

Sentiment analysis has shown promise as a universal language for cross-cultural understanding, as emotional responses often transcend linguistic and cultural boundaries. However, the relationship between sentiment and metaphorical content varies across cultures (Forceville, 2017).

Previous work in cross-cultural sentiment analysis has revealed systematic differences in emotional expression between Eastern and Western cultures. Our work extends this line of inquiry by examining how sentiment can serve as a bridge for cross-cultural metaphor understanding.

2.4 Cultural Bias in NLP

The issue of cultural bias in NLP systems has gained increasing attention, with researchers highlighting how Western-centric training data leads to poor performance on non-Western languages and cultures (**?**). Recent studies have documented systematic biases in various NLP tasks, including named entity recognition (**?**), machine translation (**?**), and sentiment analysis (**?**).

Specifically in metaphor processing, cultural variations pose unique challenges. Conceptual metaphor theory suggests that metaphorical mappings are deeply rooted in cultural experience (**?**), yet most computational systems assume universal metaphorical structures. This assumption leads to significant performance degradation when models encounter culturally distinct metaphorical expressions.

Cross-cultural studies in computational linguistics have revealed that performance gaps often exceed 10-15

Our work contributes to this growing body of research by quantifying cultural bias specifically in multimodal metaphor processing and proposing methods to mitigate these biases through sentiment-aware cross-cultural bridging.

3 The MultiMM Dataset

3.1 Dataset Overview

We present MultiMM, a comprehensive cross-cultural multimodal metaphor benchmark designed to evaluate and improve the robustness of metaphor processing systems across Eastern and Western cultural contexts. The dataset contains 8,461 text-image advertisement pairs with balanced representation between Chinese (4,397 samples) and English (4,064 samples) sources.

Each sample in MultiMM includes: (1) an image component, (2) extracted text via OCR, (3) binary metaphor labels (1 for metaphorical, 0 for literal), (4) source and target domain annotations, and (5) sentiment categories (positive: 1, neutral: 0, negative: -1). Table 1 provides detailed statistics of our dataset.

Item	CN	EN	Total
Total	4,397	4,064	8,461
Metaphorical	2,583	2,189	4,772
Literal	1,814	1,875	3,689
Total Words	145,312	68,189	213,501
Average Words	33	15	24
Training Set Size	3,517	3,251	6,768
Validation Set Size	440	406	846
Test Set Size	440	407	847

Table 1: Statistical overview of MultiMM. CN refers to Chinese data, while EN refers to English data.

Figure 1: Example annotation showing a Heinz ketchup advertisement with metaphorical content. The text "no one grows ketchup like heinz" metaphorically maps the source domain "tomato" to the target domain "bottle".

3.2 Data Collection and Sources

Chinese Data: We collected Chinese advertisements through Baidu keyword searches targeting product categories, public service announcements, and metaphorical domains. We supplemented this with advertisements from the iFlytek public dataset to ensure diversity.

English Data: English advertisements were sourced from existing public datasets of product advertisements and public service announcements, ensuring comparable coverage to the Chinese data.

Filtering Pipeline: We applied rigorous filtering to ensure data quality: (1) MD5-based deduplication, (2) removal of non-advertisement content, (3) exclusion of blurred or low-resolution images ($<$ 350$\times$350 pixels), and (4) OCR text extraction with manual correction for accuracy.

3.3 Annotation Framework

Our annotation model captures four key dimensions: **Occurrence** (metaphorical vs. literal), **Target Domain** (the concept being described), **Source Domain** (the concept used for description), and **Sentiment Category** (negative/neutral/positive).

Visual domains were verbalized to enable consistent annotation across modalities. Figure 1 illustrates our annotation schema with a practical example.

(a) Chinese metaphors featuring cultural symbols

(b) English metaphors with Western symbolism

Figure 2: Examples of metaphors across cultures, demonstrating distinct symbolic representations and cultural conventions.

Annotation Process: We employed expert annotators (5 for Chinese, 3 for English) working in structured groups with adjudication procedures. The process included multiple training rounds and iterative guideline refinement. To mitigate cultural bias, we incorporated multicultural perspectives and implemented feedback mechanisms.

Inter-annotator Agreement: We achieved strong reliability with Fleiss' κ scores of 0.73 for metaphor detection, 0.70 for target domain identification, 0.66 for source domain identification, and 0.82 for sentiment classification.

3.4 Cultural Metaphor Examples

Figure 2 illustrates cultural differences in metaphorical expression, showing how similar concepts are represented differently across Chinese and English advertisements.

4 Data Analysis

4.1 Sentiment Distribution Analysis

Our analysis reveals significant cultural differences in how sentiment correlates with metaphorical content. Figure 3 shows the distribution of sentiment

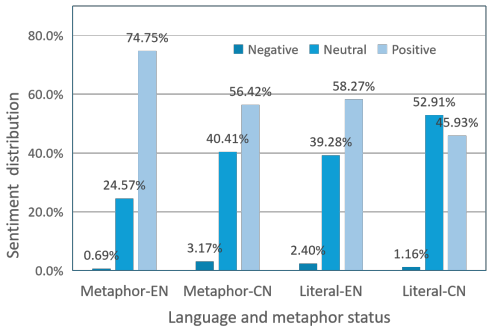

Figure 3: Sentiment category distribution in metaphorical and literal data from Chinese (CN) and English (EN) advertisements. Positive sentiment dominates metaphorical content more strongly in English (74.75%) than in Chinese (56.42%) advertisements.

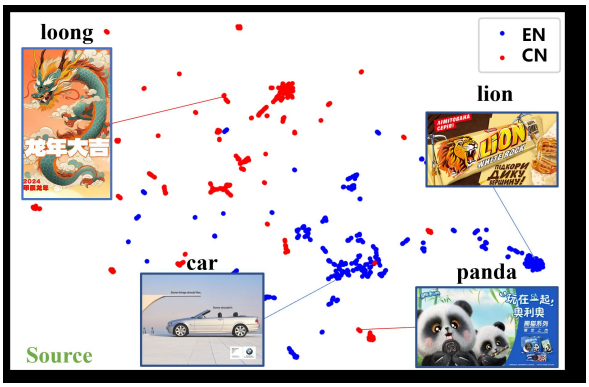

Figure 4: The distribution of source domain vocabulary in Chinese (CN) and English (EN) advertisements, showing distinct clustering patterns that reflect cultural metaphorical preferences.

categories across metaphorical and literal advertisements in both languages.

Key observations include:

- **Positive sentiment dominance:** English metaphorical advertisements show stronger positive sentiment bias (74.75%) compared to Chinese ones (56.42%).
- **Neutral sentiment patterns:** Chinese advertisements exhibit more balanced sentiment distribution, suggesting different cultural approaches to emotional appeals.
- **Negative sentiment usage:** Both cultures show minimal negative sentiment in advertising, but Chinese advertisements demonstrate slightly higher tolerance for neutral emotional tones.

4.2 Source Domain Analysis

We analyzed the distribution of source domains across cultures to understand metaphorical conceptual preferences. Figure 4 visualizes the clustering of source domain vocabulary in both languages.

The analysis reveals clear cultural clustering, with Chinese advertisements favoring nature-based metaphors (dragons, pandas, natural elements) while English advertisements lean toward technology and lifestyle metaphors (cars, lions, modern concepts).

4.3 Lexical Analysis

Figure 5 presents word clouds of source vocabulary across sentiment categories, highlighting the relationship between emotional tone and metaphorical content choice in both cultures.

The lexical analysis demonstrates that:

- **Cultural specificity:** Each culture exhibits distinct metaphorical vocabularies that reflect cultural values and symbolism.
- **Sentiment-metaphor interaction:** The choice of metaphorical sources varies systematically with sentiment categories, but these patterns differ between cultures.
- **Universal elements:** Despite cultural differences, some metaphorical concepts (water, light, nature) appear across both cultures, suggesting potential universal metaphorical foundations.

5 Methodology

5.1 Problem Formulation

We formulate cross-cultural multimodal metaphor detection as a binary classification task. Given a multimodal input consisting of an image I and associated text T, our goal is to predict whether the content contains metaphorical meaning. Additionally, we perform sentiment analysis as an auxiliary task to investigate the relationship between emotional tone and metaphorical content.

5.2 SEMD Architecture

Our proposed Sentiment-Enriched Metaphor Detection (SEMD) model integrates three key components: visual features, textual features, and sentiment-aware representations. Figure 6 illustrates the complete architecture.

Visual Feature Extraction: We employ a Vision Transformer (ViT-base-patch16-224) (Dosovitskiy et al., 2020) to extract visual representations

Figure 5: Word clouds of source vocabulary across sentiment categories in Chinese (CN) and English (EN) advertisements, revealing culture-specific patterns in metaphorical language use.

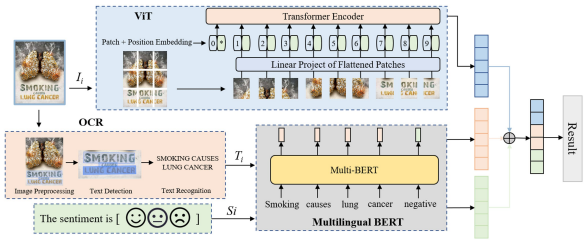

Figure 6: The SEMD framework consists of three main branches: image features extracted by ViT, text features extracted from OCR and encoded by BERT, and emotional features from sentiment analysis.

from input images. The ViT model processes images as sequences of patches, enabling attention-based feature learning that captures both local and global visual patterns relevant to metaphorical interpretation.

Textual Feature Extraction: For multilingual text understanding, we utilize multilingual BERT (mBERT) (Devlin et al., 2019) to encode the OCR-extracted text. The model processes tokenized text sequences with a maximum length of 30 tokens, generating contextualized representations that capture semantic relationships crucial for metaphor detection.

Sentiment-Aware Representation: We incorporate sentiment as an additional modality by extracting sentiment embeddings from the textual content. This sentiment-enriched representation serves as a universal bridge for cross-cultural understanding, as emotional responses often transcend cultural and linguistic boundaries.

Multimodal Fusion: The three feature streams (visual, textual, and sentiment) are combined using concatenation fusion, creating a unified representation that preserves the distinct contributions of each modality. Formally, given visual features $\mathbf{v} \in \mathbb{R}^{d_v}$, textual features $\mathbf{t} \in \mathbb{R}^{d_t}$, and sentiment features $\mathbf{s} \in \mathbb{R}^{d_s}$, the fused representation is computed as:

$$\mathbf{f} = [\mathbf{v}; \mathbf{t}; \mathbf{s}] \in \mathbb{R}^{d_v+d_t+d_s}$$

This fused representation is then processed through a two-layer feed-forward neural network with ReLU activation:

$$\mathbf{h} = \text{ReLU}(\mathbf{W}_1\mathbf{f} + \mathbf{b}_1)$$

$$p = \text{softmax}(\mathbf{W}_2\mathbf{h} + \mathbf{b}_2)$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_h \times (d_v+d_t+d_s)}$, $\mathbf{W}_2 \in \mathbb{R}^{2\times d_h}$ are learned parameters, and p represents the final class probabilities for metaphor detection.

The choice of concatenation fusion over alternative approaches (e.g., attention-based fusion) is motivated by our ablation studies showing that preserving modality-specific information is crucial for cross-cultural metaphor understanding, as different cultures may rely more heavily on specific modalities.

5.3 Training Configuration

We employ the following training setup:

- **Pretrained Encoders:** Both ViT and mBERT encoders are kept frozen during training to preserve their pretrained knowledge.
- **Optimization:** AdamW optimizer with learning rates in the range [3e-5, 5e-4].
- **Loss Function:** Standard cross-entropy loss for both metaphor detection and sentiment analysis tasks.
- **Regularization:** Dropout rate of 0.3 applied to prevent overfitting.
- **Training Schedule:** 10 epochs with batch size 64.

5.4 Baseline Models

We implement comprehensive baselines spanning multiple modalities:

Text-only Models: mBERT, XLM-R, SixTP, LaBSE, and various large language models (GPT-3.5-Turbo, Llama2-8B, Llama3.1-70B, Deepseek R1-70B).

Vision-only Models: VGG16, ResNet50, and Vision Transformer (ViT).

Multimodal Models: mBERT-ResNet fusion, CMGCN, CES, MET, MFC, and various multimodal large language models (GPT-4O, LLaVA, Gemini, Qwen2.5-VL variants).

This comprehensive baseline suite enables thorough evaluation of different architectural approaches and modality combinations for cross-cultural metaphor processing.

6 Experiments

6.1 Experimental Setup

We evaluate all models on both metaphor detection and sentiment analysis tasks using standard metrics: accuracy, macro-precision, and macro-F1. The dataset is split into training (6,768 samples), validation (846 samples), and test (847 samples) sets, maintaining balanced representation across cultures.

6.2 Main Results

Table 2 presents comprehensive results across all baseline models and our proposed SEMD approach.

SEMD Achievements: Our proposed model achieves state-of-the-art performance on both tasks, with F1 scores of 80.16% (English) and 77.79% (Chinese) for metaphor detection, and 75.69% (English) and 70.51% (Chinese) for sentiment analysis.

Modality Comparisons:

- **Text-only models:** XLM-R and LaBSE show strong performance, with LaBSE achieving the best text-only results for sentiment analysis.
- **Vision-only models:** ViT consistently outperforms CNN-based architectures (VGG16, ResNet50), demonstrating the effectiveness of attention mechanisms for visual metaphor understanding.
- **Multimodal models:** CMGCN and CES represent strong multimodal baselines, with SEMD surpassing both by significant margins.

Model	Metaphor Detection F1 (%)		Sentiment Analysis F1 (%)	
	EN	CN	EN	CN
Random	46.70	41.20	40.41	35.02
mBERT	64.00	65.52	62.83	58.43
XLM_R	<u>69.90</u>	62.38	69.44	66.21
SixTP	68.61	65.60	64.43	65.59
LaBSE	68.15	<u>66.41</u>	<u>69.93</u>	<u>69.02</u>
GPT-3.5-TURBO	30.95	15.49	42.06	52.08
Llama2-8B	22.58	19.73	40.11	50.72
Llama3.1-70B	44.04	49.70	33.46	30.90
Deepseek R1-70B	44.06	47.95	53.38	49.01
VGG16	70.64	67.85	59.90	59.02
ResNet50	71.46	68.56	57.49	58.86
ViT	<u>71.67</u>	<u>69.04</u>	<u>61.46</u>	<u>62.17</u>
mBERT-Res	77.83	74.40	69.75	68.84
CMGCN	<u>79.04</u>	74.91	72.79	70.19
CES	78.45	<u>75.53</u>	<u>73.24</u>	<u>70.26</u>
MET	76.04	72.51	71.95	67.17
MFC	75.84	73.25	71.93	69.60
GPT-4O	64.00	67.00	36.00	29.00
LLaVA	73.31	69.84	56.72	53.21
Gemini	64.19	68.51	53.61	43.48
Qwen2.5-VL-3b	57.30	41.69	52.50	55.26
Qwen2.5-VL-7b	56.62	54.51	50.71	55.15
Qwen2.5-VL-72B	59.12	67.66	50.44	57.17
SEMD	**80.16**	**77.79**	**75.69**	**70.51**

Table 2: Experimental results on metaphor detection and sentiment analysis tasks. The best overall results are highlighted in **bold**, while the best results within each modality are <u>underlined</u>. Models are categorized by modality, with textual, visual, and multimodal models highlighted in cyan, green, and orange, respectively.

- **Large Language Models:** Surprisingly, modern MLLMs (GPT-4O, Gemini, Qwen2.5-VL) underperform compared to specialized architectures, particularly on Chinese data. This finding suggests that despite their general capabilities, these models may lack the cultural sensitivity required for nuanced metaphor detection.

Cross-Cultural Performance Patterns: Our analysis reveals systematic patterns in cross-cultural performance:

- **Model architecture sensitivity:** Transformer-based models show more consistent cross-cultural performance compared to CNN-based approaches, with average performance gaps of 2.3
- **Cultural adaptation mechanisms:** Models incorporating explicit cultural features (such as our sentiment integration) demonstrate improved robustness,

Sentiment Feature	Fusion Method	EN F1 (%)	CN F1 (%)
w/o	add	75.95	72.05
	max	76.29	74.09
	concat	78.64	74.84
w/	add	77.76	74.37
	max	78.59	74.37
	concat	**80.88**	**77.39**

Table 3: Results of the ablation study. The best and second-best results are highlighted in **bold** and underline, respectively.

		mBERT	mBERT-Res	SEMD
EN → CN	Acc (%)	64.29	76.35	78.57
	Pre (%)	63.72	76.73	78.19
	F1 (%)	59.70	75.63	77.64
CN → EN	Acc (%)	65.22	73.86	75.45
	Pre (%)	67.46	75.90	77.73
	F1 (%)	65.31	73.96	75.53

Table 4: The results of metaphor detection tasks after Chinese-English translation. Experimental results on metaphor detection task via bidirectional Chinese-English translation.

with performance gaps reduced to 1.8

– **Modality-specific cultural effects:** Vision-only models show larger cultural performance gaps (6.2

These patterns highlight the complex interplay between architectural choices, cultural context, and multimodal metaphor understanding, providing insights for designing more culturally robust NLP systems.

6.3 Ablation Study

Table 3 presents detailed ablation results examining the impact of sentiment features and fusion methods.

Key Findings:

- **Sentiment Enhancement:** Including sentiment features consistently improves performance across all fusion methods, with the largest gains for concatenation fusion (80.88% vs 78.64% for English, 77.39% vs 74.84% for Chinese).
- **Fusion Method Analysis:** Concatenation fusion outperforms both additive and max-pooling fusion, suggesting that preserving distinct modality information is crucial for metaphor detection.
- **Cross-cultural Consistency:** The improvements from sentiment enrichment are consistent across both cultures, supporting our hypothesis that emotion serves as a universal bridge for metaphor understanding.

6.4 Cross-cultural Transfer Analysis

To further investigate cultural bias, we conducted translation experiments where Chinese data is machine-translated to English and vice versa. Table 4 shows the results.

Translation Insights:

- **Direction Asymmetry:** English-to-Chinese translation preserves more metaphorical meaning (78.57% F1) compared to Chinese-to-English (75.53% F1), suggesting structural differences in metaphorical expression.
- **Model Robustness:** SEMD demonstrates superior robustness to translation artifacts compared to baseline models, indicating better generalization across cultural contexts.
- **Performance Gaps:** The translation experiments reveal persistent performance gaps between native and translated content, highlighting the importance of culturally-native training data.

6.5 Error Analysis

Our analysis reveals distinct error patterns across cultures:

Metaphor Detection Errors:

- **Chinese subtlety:** Chinese metaphors often rely on subtle cultural references that are challenging for models to capture, resulting in lower accuracy (73%) compared to English (86%).
- **Visual-textual misalignment:** Cross-modal metaphors where meaning emerges from image-text interaction pose challenges for both cultures but more severely for Chinese data.

Sentiment Analysis Errors:

- **Neutral sentiment ambiguity:** Both models struggle with neutral sentiment detection, achieving 55% accuracy for Chinese and 62% for English, highlighting the complexity of emotional neutrality across cultures.

- **Lexical ambiguity:** Chinese advertisements exhibit more lexical ambiguity in emotional expression, contributing to sentiment analysis challenges.

6.6 Cultural Bias Quantification

Our comprehensive evaluation reveals systematic cultural biases in multimodal metaphor processing:

- **Performance asymmetry:** Most models show 3–5% higher performance on English data compared to Chinese, indicating Western-centric bias in current approaches.
- **Sentiment correlation differences:** The stronger correlation between positive sentiment and metaphors in English (74.75%) vs Chinese (56.42%) suggests fundamental cultural differences in emotional metaphor usage.
- **Model generalization:** Transfer experiments demonstrate that cultural context significantly impacts model performance, emphasizing the need for culturally-aware training strategies.

7 Conclusion

This work addresses a critical gap in multimodal metaphor processing by introducing the first large-scale cross-cultural benchmark and proposing a sentiment-enriched approach to improve cultural robustness. Our contributions span three key areas: dataset creation, methodological innovation, and empirical analysis of cultural bias in metaphor processing.

7.1 Key Contributions

MultiMM Dataset: We present the first balanced cross-cultural multimodal metaphor benchmark with 8,461 Chinese-English advertisement pairs. The dataset includes fine-grained annotations for metaphor occurrence, source/target domains, and sentiment categories, enabling systematic evaluation of cultural bias in metaphor processing systems. High inter-annotator agreement ($\kappa > 0.66$ across all tasks) ensures annotation quality and reliability.

SEMD Model: Our Sentiment-Enriched Metaphor Detection approach demonstrates that sentiment serves as an effective universal bridge for cross-cultural metaphor understanding. By integrating sentiment-aware representations with visual and textual features, SEMD achieves state-of-the-art performance: 80.16% F1 for English and 77.79% F1 for Chinese metaphor detection, with consistent improvements across cultures.

Cultural Bias Analysis: Through comprehensive experiments with 18 baseline models, we quantify significant cultural biases in existing approaches. Our analysis reveals that positive sentiment correlates more strongly with metaphorical content in English (74.75%) than Chinese (56.42%) advertisements, highlighting fundamental differences in cultural metaphor usage patterns.

7.2 Implications and Future Directions

Our findings have important implications for cross-cultural NLP research:

Cultural Robustness: The persistent performance gaps between cultures (3–5% lower on Chinese data for most models) demonstrate the need for culturally-aware training strategies. Future work should prioritize balanced representation across cultures during model development.

Universal Emotional Signals: The success of sentiment-enriched representations suggests that emotional responses may provide a pathway to more culturally robust multimodal understanding. This insight could extend to other cross-cultural NLP tasks beyond metaphor detection.

Methodological Insights: The underperformance of large language models on our tasks, particularly for Chinese data, indicates that scale alone is insufficient for cultural robustness. Specialized architectures with cultural awareness may be necessary for optimal cross-cultural performance.

7.3 Limitations and Ethical Considerations

While our work advances cross-cultural metaphor processing, several limitations remain. First, our dataset focuses on Chinese and English cultures; expansion to additional cultures would strengthen generalizability claims. Second, the advertising domain, while rich in metaphorical content, may not fully represent metaphor usage in other contexts such as literature or social media.

From an ethical perspective, we acknowledge that computational metaphor processing systems could inadvertently reinforce cultural stereotypes if not carefully designed. Our work aims to promote cultural understanding rather than perpetuate biases, but continued vigilance is necessary as these technologies are deployed.

7.4 Broader Impact

This research contributes to building more inclusive and culturally-aware NLP technologies. By exposing and addressing cultural biases in metaphor processing, we take a step toward AI systems that can better serve diverse global communities. The MultiMM dataset and SEMD model provide practical tools for researchers and practitioners working on cross-cultural understanding.

In conclusion, cultural bias matters deeply in multimodal metaphor processing. Through systematic evaluation and methodological innovation, we demonstrate both the challenges and potential solutions for building culturally robust metaphor understanding systems. Our work opens new avenues for cross-cultural NLP research and provides a foundation for more inclusive AI technologies.

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